Technical Note: block-VIEM.jl

Volume Integral Equation Method with SWG/half-SWG Basis,
Duffy Singular Integration, and AIM-FFT Acceleration
for Light Scattering by Arbitrary Dielectric Particles

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Contents

1 Introduction

The volume integral equation method (VIEM) computes the electromagnetic field scattered by an inhomogeneous dielectric particle of arbitrary shape by solving an integral equation on the particle volume rather than on its surface [1, 2]. Compared with the discrete dipole approximation (DDA) [5, 6, 7] implemented in the companion package `block-DDA_Py` [9], a volumetric formulation with conforming vector basis functions on a tetrahedral mesh offers two practical advantages: (i) shapes with curved or faceted surfaces are represented without stair-casing, and (ii) the number of degrees of freedom per wavelength required for a given accuracy is considerably smaller than for a cubic-lattice DDA, especially for high refractive index contrast.

This document describes the theoretical background and the algorithms implemented in `block-VIEM.jl` v0.1.2, covering:

1. the EFVIE-D formulation with Schaubert–Wilton–Glisson (SWG) and half-SWG basis functions (Sec. 2),
2. weakening the $1/R^3$ singularity to $1/R$ via integration by parts, including the boundary-face surface-correction terms K^B, K^C, K^D required to restore charge neutrality (Sec. 3),
3. evaluation of the remaining $1/R$ singular and near-singular volume integrals via the generalized Duffy transform (Sec. 4),
4. FFT-accelerated matrix–vector products through the Adaptive Integral Method (AIM) with moment matching and precorrection (Sec. 5),
5. the direct and Krylov iterative solvers and the multi-orientation batch solve (Sec. 6),
6. the CAS-v2 compatible scattering observables computed from the solution (Sec. 7),
7. validation against Mie theory and the oblate-spheroid symmetry identity (Sec. 8),
8. the HDF5 output format for spheroid parameter sweeps, bit compatible with the `block-DDA_Py` schema consumed by `build_spheroid_lut.py` (Sec. 9).

Throughout this note the electromagnetic equations are written in the physics (BH83 [8]) time convention $\mathrm{e}^{-i\omega t}$, so that an outgoing scalar Helmholtz Green's function carries the factor $\mathrm{e}^{+ik_0R}/(4\pi R)$. Absorbing materials have $\mathrm{Im}(\epsilon_p) > 0$. Lengths are expressed in micrometres (µm). Vectors are set in bold italic ($\boldsymbol{E}$, $\boldsymbol{k}$, $\boldsymbol{r}$), matrices and second-rank tensors in upright bold ($\mathbf{A}$, $\mathbf{G}$), and unit vectors carry a hat ($\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{n}}$).

2 VIEM Formulation

2.1 EFVIE-D strong form

Consider a non-magnetic ($\mu = \mu_0$) inhomogeneous dielectric particle occupying the volume Ω with complex permittivity $\epsilon_p(\boldsymbol{r})$, embedded in a homogeneous background of real permittivity ϵ_{bg}. The wavenumber in the background is

$$k_0 = \omega\,\sqrt{\mu_0\,\epsilon_{\mathrm{bg}}} = \frac{2\pi\, m_m}{\lambda_0}, \tag{1}$$

where $m_m = \sqrt{\epsilon_{\mathrm{bg}}/\epsilon_0}$ is the refractive index of the background medium.

We use the electric flux volume integral equation in the *D-form* [1]: the unknown field is the electric flux density $\boldsymbol{D}(\boldsymbol{r}) = \epsilon_p(\boldsymbol{r})\,\boldsymbol{E}(\boldsymbol{r})$, whose normal component is continuous across material interfaces, so the scheme places all dielectric jumps inside the basis functions and not in the expansion coefficients. Define the dielectric contrast

$$\kappa(\boldsymbol{r}) = \frac{\epsilon_p(\boldsymbol{r}) - \epsilon_{\mathrm{bg}}}{\epsilon_p(\boldsymbol{r})}. \tag{2}$$

The equivalent volume polarisation current is $\boldsymbol{J}_v = -i\omega\,\kappa\,\boldsymbol{D}$, and the EFVIE-D strong form reads [1, 2]

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{r}) = \frac{\boldsymbol{D}(\boldsymbol{r})}{\epsilon_p(\boldsymbol{r})} - \frac{1}{\epsilon_{\mathrm{bg}}}\big(k_0^2 + \boldsymbol{\nabla}\boldsymbol{\nabla}\cdot\big)\int_\Omega \kappa(\boldsymbol{r}')\,\boldsymbol{D}(\boldsymbol{r}')\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^3r', \tag{3}$$

where $G(\boldsymbol{r},\boldsymbol{r}')$ is the free-space scalar Helmholtz Green's function

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{\mathrm{e}^{+ik_0R}}{4\pi R}, \qquad R = \big|\boldsymbol{r} - \boldsymbol{r}'\big|. \tag{4}$$

2.2 Incident plane wave

The incident field is a monochromatic plane wave

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{r}) = \boldsymbol{E}_0\,\exp\!\big(+ik_0\,\hat{\boldsymbol{k}}_{\mathrm{inc}}\cdot\boldsymbol{r}\big), \tag{5}$$

with $\hat{\boldsymbol{k}}_{\mathrm{inc}}$ the unit propagation direction and $\boldsymbol{E}_0$ a complex polarisation vector perpendicular to $\hat{\boldsymbol{k}}_{\mathrm{inc}}$. For CAS-v2 observables the incident polarisation is left-handed circular (matching `block-DDA_Py` [9]):

$$\boldsymbol{E}_0 = \frac{1}{\sqrt{2}}\,\hat{\boldsymbol{e}}_{\theta,\mathrm{inc}} + \frac{i}{\sqrt{2}}\,\hat{\boldsymbol{e}}_{\phi,\mathrm{inc}}, \tag{6}$$

where $\hat{\boldsymbol{e}}_{\theta,\mathrm{inc}}$, $\hat{\boldsymbol{e}}_{\phi,\mathrm{inc}}$ are the polar and azimuthal unit vectors of the incident direction in the particle frame. Particle orientation is parameterised by intrinsic ZYZ Euler angles (α,β,γ), following `scipy.spatial.transform.Rotation` and `block-DDA_Py`. We denote by

$$\mathbf{R}(\alpha,\beta,\gamma) \;=\; \mathbf{R}_z(\alpha)\,\mathbf{R}_y(\beta)\,\mathbf{R}_z(\gamma) \tag{7}$$

the active rotation that maps particle-frame coordinates of a physical vector to its lab-frame coordinates; equivalently, (7) is the fixed-frame composition obtained by applying the three extrinsic rotations $\mathbf{R}_z(\gamma)$, $\mathbf{R}_y(\beta)$, $\mathbf{R}_z(\alpha)$ in that order about the lab axes. The laboratory-frame triad $\{\hat{\boldsymbol{u}}_{\mathrm{inc},L}, \hat{\boldsymbol{e}}_{\theta,L}, \hat{\boldsymbol{e}}_{\phi,L}\}$ is therefore expressed in the particle frame by applying $\mathbf{R}(\alpha,\beta,\gamma)^\top$ $(= \mathbf{R}(\alpha,\beta,\gamma)^{-1}$ since $\mathbf{R}$ is orthogonal).

2.3 SWG and half-SWG basis functions

The particle volume Ω is discretised into a conforming tetrahedral mesh $\mathcal{T}_h$ with N_{tet} elements. On each internal face S_n shared by two tetrahedra T_n^+ and T_n^-, an SWG basis function [1, Eqs. (10), (12)] is defined by

$$\boldsymbol{f}_n(\boldsymbol{r}) = \begin{cases} \dfrac{a_n}{3\,V_n^+}\,\big(\boldsymbol{r} - \boldsymbol{p}_n^+\big), & \boldsymbol{r} \in T_n^+, \\ -\dfrac{a_n}{3\,V_n^-}\,\big(\boldsymbol{r} - \boldsymbol{p}_n^-\big), & \boldsymbol{r} \in T_n^-, \\ \boldsymbol{0}, & \text{otherwise}, \end{cases} \tag{8}$$

where a_n is the face area, $V_n^\pm$ the volumes of the two supporting tetrahedra, and $\boldsymbol{p}_n^\pm$ the free vertex (the vertex of $T_n^\pm$ not on S_n). The normalisation in (8) enforces $\boldsymbol{f}_n\cdot\hat{\boldsymbol{n}} = 1$ on S_n (oriented from T_n^+ to T_n^-) and zero on the other five faces of $T_n^+ \cup T_n^-$. The divergence is piecewise constant,

$$\nabla \cdot \boldsymbol{f}_n(\boldsymbol{r}) = \begin{cases} +\dfrac{a_n}{V_n^+}, & \boldsymbol{r} \in T_n^+, \\ -\dfrac{a_n}{V_n^-}, & \boldsymbol{r} \in T_n^-, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

A naive SWG expansion that uses only internal faces implicitly imposes $\boldsymbol{D}\cdot\hat{\boldsymbol{n}} = 0$ on the particle boundary $\partial\Omega$, leaving the surface polarisation charge unrepresented and producing systematic cross-section errors of order 10–30% on Mie tests. To restore charge neutrality, the original [1] formulation adds one half-SWG basis function per boundary face. On a boundary face S_n with a single supporting tetrahedron T_n^+, the half-SWG function is the first branch of (8) alone:

$$\boldsymbol{f}_n^{\text{half}}(\boldsymbol{r}) = \frac{a_n}{3\,V_n^+}\,\big(\boldsymbol{r} - \boldsymbol{p}_n^+\big), \qquad \boldsymbol{r} \in T_n^+, \tag{10}$$

with $\boldsymbol{f}_n\cdot\hat{\boldsymbol{n}} = 1$ on S_n and zero on the other three faces of T_n^+. For a half-SWG function the normal flux does *not* vanish on $\partial\Omega$, so it carries an induced surface polarisation charge $\sigma = \kappa\,\boldsymbol{D}\cdot\hat{\boldsymbol{n}}$ whose integral over S_n exactly cancels the bulk bound charge of the same basis (Sec. 3.3). The electric flux density is expanded as

$$\boldsymbol{D}(\boldsymbol{r}) \approx \sum_{n=1}^{N_{\text{dof}}} D_n\,\boldsymbol{f}_n(\boldsymbol{r}), \qquad N_{\text{dof}} = N_{\text{int. faces}} + N_{\text{bdy. faces}}. \tag{11}$$

2.4 Galerkin weak form

Testing (3) with a basis function $\boldsymbol{f}_m$ and substituting (11) yields the $N_{\text{dof}} \times N_{\text{dof}}$ linear system

$$\sum_{n=1}^{N_{\text{dof}}} Z_{mn}\,D_n = b_m, \qquad b_m = \int_\Omega \boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{E}_{\text{inc}}(\boldsymbol{r})\,\mathrm{d}^3 r, \tag{12}$$

with the Galerkin impedance matrix element

$$\begin{aligned} Z_{mn} = &\int_\Omega \frac{\boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{f}_n(\boldsymbol{r})}{\epsilon_p(\boldsymbol{r})}\,\mathrm{d}^3 r \\ &- \frac{1}{\epsilon_{\text{bg}}} \int_\Omega \boldsymbol{f}_m(\boldsymbol{r}) \cdot \Big[\big(k_0^2 + \nabla\nabla\cdot\big) \int_\Omega \kappa(\boldsymbol{r}')\,\boldsymbol{f}_n(\boldsymbol{r}')\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^3 r'\Big]\mathrm{d}^3 r. \end{aligned} \tag{13}$$

For a homogeneous particle, κ and ϵ_p are constants and factor out of the inner integral, giving the compact decomposition

$$Z_{mn} = \frac{1}{\epsilon_p}\,M_{mn} - \frac{\kappa}{\epsilon_{\text{bg}}}\,K_{mn}, \tag{14}$$

where $M_{mn} = \int \boldsymbol{f}_m\cdot\boldsymbol{f}_n\,\mathrm{d}^3 r$ is the mass matrix and K_{mn} is the radiation kernel defined below.

3 Weakening the Singularity via Integration by Parts

As written in (13), the radiation kernel $(k_0^2+\nabla\nabla\cdot)G$ contains a hyper-singular term $1/R^3$ coming from $\nabla\nabla\cdot G$. Moving the two derivatives off G onto $\boldsymbol{f}_m$ and $\boldsymbol{f}_n$ by successive integrations by parts reduces the singularity to the weakly singular $1/R$ kernel that Duffy quadrature can handle. Because SWG and half-SWG functions are only piecewise smooth and only half-SWG functions have nonzero $\boldsymbol{f}_n\cdot\hat{\boldsymbol{n}}$ on $\partial\Omega$, the by-parts procedure generates three boundary-face contributions in addition to the standard bulk term.

3.1 Bulk–bulk term K^A

Apply $\boldsymbol{\nabla\nabla\cdot}$ to the inner integral and move the inner divergence onto $\boldsymbol{f}_n$ using the divergence theorem; do the same for the outer divergence and $\boldsymbol{f}_m$. For internal SWG basis functions (vanishing normal flux on $\partial\Omega$) the boundary terms are identically zero and the resulting bulk kernel is

$$K^A_{mn} = \int_\Omega\int_\Omega \big[k_0^2\, \boldsymbol{f}_m(\boldsymbol{r})\cdot\boldsymbol{f}_n(\boldsymbol{r}') - (\boldsymbol{\nabla\cdot f}_m)(\boldsymbol{r})\,(\boldsymbol{\nabla}\boldsymbol{\cdot}'\,\boldsymbol{f}_n)(\boldsymbol{r}')\big]\, G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^3r'\,\mathrm{d}^3r. \tag{15}$$

The integrand now contains only the weakly singular $1/R$ kernel and is evaluated tetrahedron pair by tetrahedron pair.

3.2 Surface correction terms K^B, K^C, K^D

When the source basis $\boldsymbol{f}_n$ is a half-SWG function attached to a boundary face S_n, the inner integration by parts leaves a surface term from the boundary $\partial\Omega$:

$$\Phi_n(\boldsymbol{r}) \equiv \boldsymbol{\nabla\cdot}\int_\Omega G(\boldsymbol{r},\boldsymbol{r}')\,\boldsymbol{f}_n(\boldsymbol{r}')\,\mathrm{d}^3r' = \int_\Omega G\,\boldsymbol{\nabla}\boldsymbol{\cdot}'\,\boldsymbol{f}_n\,\mathrm{d}^3r' \;-\; \oint_{\partial\Omega} G(\boldsymbol{r},\boldsymbol{r}')\,\big(\boldsymbol{f}_n\!\cdot\hat{\boldsymbol{n}}'\big)\,\mathrm{d}^2r'. \tag{16}$$

Similarly, the outer integration by parts against $\boldsymbol{f}_m$ produces a surface term if $\boldsymbol{f}_m$ is a half-SWG function on a boundary face S_m. The complete kernel is

$$K_{mn} = K^A_{mn} + K^B_{mn} + K^C_{mn} + K^D_{mn}, \tag{17}$$

with the three boundary contributions

$$K^B_{mn} = \int_\Omega\oint_{\partial\Omega} (\boldsymbol{\nabla\cdot f}_m)(\boldsymbol{r})\,\big(\boldsymbol{f}_n\!\cdot\hat{\boldsymbol{n}}'\big)\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^2r'\,\mathrm{d}^3r, \tag{18}$$

$$K^C_{mn} = \oint_{\partial\Omega}\int_\Omega \big(\boldsymbol{f}_m\!\cdot\hat{\boldsymbol{n}}\big)\,(\boldsymbol{\nabla}\boldsymbol{\cdot}'\,\boldsymbol{f}_n)(\boldsymbol{r}')\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^3r'\,\mathrm{d}^2r, \tag{19}$$

$$K^D_{mn} = -\oint_{\partial\Omega}\oint_{\partial\Omega} \big(\boldsymbol{f}_m\!\cdot\hat{\boldsymbol{n}}\big)\,\big(\boldsymbol{f}_n\!\cdot\hat{\boldsymbol{n}}'\big)\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^2r'\,\mathrm{d}^2r. \tag{20}$$

By the half-SWG property $\boldsymbol{f}_n\cdot\hat{\boldsymbol{n}} = 1$ on S_n and zero on the remaining boundary faces, the surface integrals in (18)–(20) collapse to integrals over the specific faces S_m and S_n: K^B_{mn} is non-zero only when n is a boundary face, K^C_{mn} only when m is a boundary face, and K^D_{mn} only when both m and n are boundary faces.

3.3 Charge neutrality

The induced bound charge associated with a single half-SWG basis on a boundary face S_n is the sum of a bulk contribution $\rho_{\mathrm{vol}} = -\kappa\,\boldsymbol{\nabla\cdot D}$ and a surface contribution $\sigma_{\mathrm{surf}} = \kappa\,\boldsymbol{D}\!\cdot\!\hat{\boldsymbol{n}}$. Using (9) and the unit-flux property on S_n,

$$Q_{\mathrm{vol}} = -\int_{T_n^+} \kappa\,\boldsymbol{\nabla\cdot f}_n\,\mathrm{d}^3r = -\kappa\,\frac{a_n}{V_n^+}\cdot V_n^+ = -\kappa\,a_n, \tag{21}$$

$$Q_{\mathrm{surf}} = \int_{S_n} \kappa\,(\boldsymbol{f}_n\!\cdot\hat{\boldsymbol{n}})\,\mathrm{d}^2r = \kappa\,a_n. \tag{22}$$

The sum $Q_{\mathrm{vol}} + Q_{\mathrm{surf}} = 0$ holds exactly *only* when the surface contributions K^B, K^C, K^D are retained. Dropping them, as an internal-only SWG expansion does, leaves the particle with a spurious net bound charge $-\kappa\,a_n$ per boundary face and produces cross-section errors of order 10–30% on Mie tests. With the full decomposition (17), block-VIEM.jl reproduces the Mie sphere absorption cross section to $\sim 0.2\%$ at $\ell_c = 0.5$ (589 DoFs), an order-of-magnitude improvement per DoF over cubic-lattice DDA.

3.4 Mass matrix

The first term in (14) is the sparse mass matrix

$$M_{mn} = \int_{T_m \cap T_n} \boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{f}_n(\boldsymbol{r}) \, \mathrm{d}^3 r. \tag{23}$$

The integrand is a degree-2 polynomial in each tetrahedron, so the 5-point Gauss rule on the tetrahedron (degree 3 exact) evaluates M_{mn} without quadrature error.

4 Singular and Near-Singular Volume Integration

4.1 Generalised vertex Duffy transform

The weakly singular kernel $1/R$ in (15)–(20) remains integrable but standard Gauss quadrature fails when the source and test tetrahedra share a vertex, edge, or face, or when the outer Gauss point approaches a face of the inner element. Following Mousavi–Sukumar [4], all such cases are handled uniformly by a *vertex-only* Duffy transform applied to sub-tetrahedra obtained by projecting the singular point onto the inner element.

For $\alpha = 1$ (a $1/R$ singularity at a single vertex), the optimised transform from the unit cube $\boldsymbol{u} = (u_1, u_2, u_3) \in [0,1]^3$ to barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ with $\lambda_1 = 1$ at the singular vertex is [4, Eq. (1) with $\beta = 1$]

$$\lambda_1 = 1 - u_1, \quad \lambda_2 = u_1(1 - u_2), \quad \lambda_3 = u_1 u_2 (1 - u_3), \quad \lambda_4 = u_1 u_2 u_3. \tag{24}$$

The face $u_1 = 0$ of the cube collapses entirely to the singular vertex, and $R \propto u_1$ to leading order along any ray departing from it. The total Jacobian combining (24) with the affine map from reference to physical coordinates is

$$J_D(\boldsymbol{u}) = 6 \, |V_{\mathrm{tet}}| \, u_1^2 \, u_2. \tag{25}$$

The factor u_1^2 exactly cancels the $1/R$ singularity, leaving a smooth integrand on the unit cube that is integrated by a tensor-product Gauss–Legendre rule. For the K^A kernel (15) the recommended order is $5 \times 5 \times 5$ (125 points) for self and face-adjacent pairs, and a $4 \times 4 \times 4$ rule for edge- and vertex-adjacent pairs; a 5-point Gauss rule on the source tetrahedron is sufficient for all pairs that are not near the observation point.

4.2 Pair classification

Given an outer quadrature point $\boldsymbol{r}$ inside the test tetrahedron T_m, the source element T_n is classified and integrated according to the minimum distance:

- *self* ($T_n = T_m$): the outer Gauss point is the singular vertex. The source tetrahedron is subdivided into four sub-tetrahedra with the outer Gauss point as a common vertex, and each is integrated by the Duffy transform (24).
- *face-/edge-/vertex-adjacent*: the singular point is the nearest vertex, the nearest point on the shared edge, or the shared face. Sub-tetrahedra are generated as above around that singular point and integrated by the same Duffy rule.
- *far* (R larger than a chosen threshold): the inner integral is evaluated with a plain tetrahedron Gauss rule (5 points). Pairs further than the AIM near-field radius are never integrated directly (Sec. 5).

4.3 Surface integrals for K^B, K^C, K^D

The surface contributions in Sec. 3.2 require the integration of G over a triangle, optionally combined with an inner tetrahedron integral (K^B, K^C) or an outer triangle integral (K^D). We adopt a two-dimensional version of the Duffy transform on the reference triangle: from $(u_1, u_2) \in [0,1]^2$ to barycentric coordinates (η_1, η_2, η_3),

$$\eta_1 = 1 - u_1, \quad \eta_2 = u_1(1 - u_2), \quad \eta_3 = u_1 u_2, \tag{26}$$

with Jacobian $J_{D,2}(\boldsymbol{u}) = 2|A_{\text{tri}}|\, u_1$. The linear factor u_1 cancels the $1/R$ singularity at the vertex $\eta_1 = 1$, and the face $u_1 = 0$ collapses to that vertex just as in the three-dimensional case. For the K^D self/edge/vertex branches, the outer triangle is first subdivided into sub-triangles at the singular feature and the 2D Duffy rule is applied to each.

The near-singular sub-cases of K^B and K^C (for example, the outer tetrahedron T_m having a face coincident with the inner boundary face S_n) are handled by projecting the outer Gauss point onto the inner triangle and applying the 2D Duffy rule around that projection.

5 FFT-Accelerated MVP: Adaptive Integral Method

Direct computation and storage of the dense impedance matrix Z scales as $\mathcal{O}(N_{\text{dof}}^2)$ in memory and $\mathcal{O}(N_{\text{dof}}^3)$ in operations for direct factorisation, quickly becoming prohibitive. The Adaptive Integral Method (AIM) [3] accelerates the matrix–vector product by representing the far-field part of the kernel on an auxiliary cubic grid and evaluating the resulting block-Toeplitz sum with a three-dimensional FFT, mirroring the Goodman–Draine–Flatau construction [7] used by `block-DDA_Py`.

5.1 Auxiliary grid and stencil

A Cartesian grid of spacing h_g is overlaid on the bounding box of the particle. Let $\boldsymbol{g}_p$ denote the position of grid point p. Each basis function $\boldsymbol{f}_n$ is associated with a stencil $\mathcal{S}_n$ of $M_{\text{st}} = (n_{\text{st}}+1)^3$ nearby grid points surrounding its centroid. For a moment matching order n_{st}, the stencil is a cube with $n_{\text{st}} + 1$ grid points on each side.

5.2 Moment matching

For each basis function $\boldsymbol{f}_n$ and each stencil point $p \in \mathcal{S}_n$ we construct a projection weight $W_{np} \in \mathbb{C}^3$ so that the multipole moments of the point source at p match those of the continuous source $\boldsymbol{f}_n$ up to order n_{st}. The moments are defined with respect to the basis centroid $\boldsymbol{c}_n$:

$$\mu_n^{\boldsymbol{\alpha}} = \int_\Omega \boldsymbol{f}_n(\boldsymbol{r})\, (\boldsymbol{r} - \boldsymbol{c}_n)^{\boldsymbol{\alpha}}\, \mathrm{d}^3 r, \qquad |\boldsymbol{\alpha}| = \alpha_x + \alpha_y + \alpha_z \le n_{\text{st}}, \tag{27}$$

and the weights $\{W_{np}\}_{p\in\mathcal{S}_n}$ are defined by the linear constraint

$$\sum_{p\in\mathcal{S}_n} W_{np}\, (\boldsymbol{g}_p - \boldsymbol{c}_n)^{\boldsymbol{\alpha}} = \mu_n^{\boldsymbol{\alpha}}, \qquad \forall\, |\boldsymbol{\alpha}| \le n_{\text{st}}. \tag{28}$$

A companion set of weights matches the divergence moments $\int (\boldsymbol{\nabla} \cdot \boldsymbol{f}_n)(\boldsymbol{r} - \boldsymbol{c}_n)^{\boldsymbol{\alpha}}\, \mathrm{d}^3 r$ for the scalar potential part of the kernel. The linear system (28) is small (M_{st} unknowns per basis function, $\binom{n_{\text{st}}+3}{3}$ equations) and is solved once, during matrix setup, per basis function.

5.3 Far-field product via FFT

With the moment matching complete, the far-field part of the matrix–vector product $\boldsymbol{y}^{\mathrm{far}} = K^{\mathrm{far}}\,\boldsymbol{x}$ is evaluated in three steps:

1. *Projection.* For each basis function n and each stencil point p, accumulate $q_p = \sum_n W_{np}\,x_n$ onto the grid.

2. *Convolution.* Compute $Q_p = \sum_{p'} G(\boldsymbol{g}_p - \boldsymbol{g}_{p'})\,q_{p'}$ via a three-dimensional FFT of q, pointwise multiplication by the precomputed FFT of G on the doubled grid, and an inverse FFT.

3. *Interpolation.* For each test function m, $y_m^{\mathrm{far}} = \sum_{p\in\mathcal{S}_m} W_{mp}^{\top}\,Q_p$.

Because G depends only on the grid-point separation, its convolution matrix is block-Toeplitz and is computed with the 3D FFT of the same construction used in `block-DDA_Py` [7]. The cost is $\mathcal{O}(N_g \log N_g)$ per MVP, where N_g is the number of grid points. Memory for the grid Green function is also $\mathcal{O}(N_g)$ on the doubled domain.

5.4 Precorrection of near-field pairs

The FFT sum above is accurate only in the far field; for pairs with $\|\boldsymbol{c}_m - \boldsymbol{c}_n\| \le r_{\mathrm{near}}$ it is replaced by the exact integrals of Sec. 4. The construction is a *difference* of two quantities: for each near pair (m,n) we compute the exact K_{mn} and subtract from it the contribution that the FFT would have produced through the grid projection, storing the difference in a sparse precorrection matrix K^{pc}. The complete MVP is

$$\boldsymbol{y} = \left(D_M - \tfrac{\kappa}{\epsilon_{\mathrm{bg}}}(K^{\mathrm{far}} + K^{\mathrm{pc}})\right)\boldsymbol{x}, \tag{29}$$

where D_M is the sparse diagonal mass contribution M/ϵ_p and K^{pc} has $\mathcal{O}(N_{\mathrm{dof}})$ non-zero entries. The default near radius is $r_{\mathrm{near}} = 2\times$ (stencil extent), verified in Phase-3 benchmarks to produce $< 0.5\%$ relative error between the AIM-accelerated and the dense matrix–vector products.

6 Solvers

6.1 Direct and Krylov solvers

For small problems ($N_{\mathrm{dof}} \lesssim$ a few thousand), the dense assembly of Z and its LU factorisation is practical and serves as a validation reference for the AIM path. For larger problems we use a Krylov iterative method (BiCGSTAB from `Krylov.jl` [10]) applied to the AIM operator (29). A Jacobi preconditioner formed from the diagonal of the sparse mass and precorrection contributions suffices in our tests.

6.2 Multi-orientation batch solve

Because the impedance matrix Z depends only on the particle geometry and permittivity, it can be assembled and factorised once and then reused across many orientations: each orientation changes only the right-hand side $\boldsymbol{b}$. The function `solve_cas_v2_orientations` assembles Z, computes its LU factorisation, and then back-substitutes for every $(\alpha_\ell, \beta_\ell, \gamma_\ell)$ in the input list, returning one `CASv2Result` per orientation. For spheroids this is particularly efficient: only the $\alpha = 0$ orientation needs an independent solve for each β, and the remaining α values are obtained by the analytical expansion of Sec. 8.2.

7 Scattering Observables

Let D_n ($n = 1, \ldots, N_{\mathrm{dof}}$) denote the solution of (12) and $\boldsymbol{D}(\boldsymbol{r}) = \sum D_n \, \boldsymbol{f}_n(\boldsymbol{r})$ the reconstructed electric flux density. This section collects the observables computed from D_n that correspond to the CAS-v2 detector protocol [9] and can be compared directly with `block-DDA_Py`.

7.1 Far-field scattering amplitude

The scattered far field in the physics convention reads

$$\boldsymbol{E}^{\mathrm{sca}}(\boldsymbol{r}) \;\sim\; \frac{\mathrm{e}^{+ik_0 r}}{4\pi\, r}\, \boldsymbol{F}(\hat{\boldsymbol{r}}), \qquad r \to \infty, \tag{30}$$

where the vector scattering amplitude is obtained by transverse projection of the Fourier transform of $\boldsymbol{D}$:

$$\boldsymbol{F}(\hat{\boldsymbol{r}}) = \frac{k_0^2\, \kappa}{\epsilon_{\mathrm{bg}}} \big(\mathbf{I}_3 - \hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}\big) \cdot \sum_{n=1}^{N_{\mathrm{dof}}} D_n \int_{\Omega} \boldsymbol{f}_n(\boldsymbol{r}')\, \mathrm{e}^{-ik_0\, \hat{\boldsymbol{r}} \cdot \boldsymbol{r}'}\, \mathrm{d}^3 r'. \tag{31}$$

The standard scattering amplitude used in Bohren–Huffman [8] is $\boldsymbol{f}(\hat{\boldsymbol{r}}) = \boldsymbol{F}(\hat{\boldsymbol{r}})/(4\pi)$.

7.2 Absorption cross section

The absorption cross section is evaluated directly from the volumetric Joule-loss formula, which is bilinear in $\boldsymbol{D}$ and hence reduces to a quadratic form in the expansion coefficients:

$$C_{\mathrm{abs}} = \frac{k_0\ \mathrm{Im}(\epsilon_p)}{\epsilon_{\mathrm{bg}}\, |\epsilon_p|^2\, |\boldsymbol{E}_0|^2}\ \mathrm{Re}\big(\boldsymbol{D}^\dagger \mathbf{M}\, \boldsymbol{D}\big), \tag{32}$$

where $\mathbf{M}$ is the mass matrix (23). For real ϵ_p this identically vanishes.

7.3 Scattering cross section by far-field integration

The scattering cross section is computed as the integral of the radiated flux density over the unit sphere,

$$C_{\mathrm{sca}} = \frac{1}{16\pi^2\, |\boldsymbol{E}_0|^2} \int_{S^2} |\boldsymbol{F}(\hat{\boldsymbol{r}})|^2 \, \mathrm{d}\Omega, \tag{33}$$

using a product Gauss–Legendre (in $\cos\theta$) times trapezoid (in ϕ) rule with $2n_\theta^2$ directions. The default $n_\theta = 10$ is sufficient for size parameters $k_0\, r_{ve} \lesssim 5$; larger targets use $n_\theta \geq 20$. The quadratic functional $|\boldsymbol{F}|^2$ is nonnegative and insensitive to the sign-convention subtleties that plague forward-direction optical-theorem estimates of C_{ext}, so in practice we obtain C_{ext} from

$$C_{\mathrm{ext}} = C_{\mathrm{abs}} + C_{\mathrm{sca}}. \tag{34}$$

7.4 PCAS forward amplitude

The CAS-v2 forward scattering observable is defined with a circular incident polarisation [9]. Let $\hat{\boldsymbol{u}}_{\mathrm{fw}}$ be the forward scattering direction (equal to the incident propagation direction) and $\hat{\boldsymbol{e}}_{\theta,\mathrm{fw}}, \hat{\boldsymbol{e}}_{\phi,\mathrm{fw}}$ the associated polar and azimuthal unit vectors in the particle frame. The s- and p-channel CAS observables are

$$S^{\mathrm{PCAS}}_{\mathrm{fw},\theta}\ [= S_s(0)] = \frac{\sqrt{2}}{4\pi}\, \boldsymbol{F}(\hat{\boldsymbol{u}}_{\mathrm{fw}}) \cdot \hat{\boldsymbol{e}}_{\theta,\mathrm{fw}}, \tag{35}$$

$$S^{\mathrm{PCAS}}_{\mathrm{fw},\phi}\ [= S_p(0)] = -\,\frac{i\,\sqrt{2}}{4\pi}\, \boldsymbol{F}(\hat{\boldsymbol{u}}_{\mathrm{fw}}) \cdot \hat{\boldsymbol{e}}_{\phi,\mathrm{fw}}, \tag{36}$$

where the $\sqrt{2}$ and $i\sqrt{2}$ factors are inherited from the circular polarisation decomposition (6), and the $1/(4\pi)$ factor converts from $\boldsymbol{F}$ to the BH83-normalised scattering amplitude $\boldsymbol{f} = \boldsymbol{F}/(4\pi)$. The PCAS forward observable reported to the downstream CAS-v2 analyser is

$$S_{\mathrm{fw}}^{\mathrm{PCAS}} = \frac{S_{\mathrm{fw},\theta}^{\mathrm{PCAS}} + S_{\mathrm{fw},\phi}^{\mathrm{PCAS}}}{2}. \tag{37}$$

7.5 OCBS backward amplitude

For the exact backward scattering direction $\hat{\boldsymbol{u}}_{\mathrm{bk}} = -\hat{\boldsymbol{u}}_{\mathrm{inc}}$, the laboratory basis is $\hat{\boldsymbol{e}}_{\theta,\mathrm{bk}} = -\hat{\boldsymbol{e}}_{\theta,\mathrm{inc}}$ and $\hat{\boldsymbol{e}}_{\phi,\mathrm{bk}} = +\hat{\boldsymbol{e}}_{\phi,\mathrm{inc}}$. The θ- and ϕ-channel backward amplitudes are defined in direct analogy with (35)–(36) at $\hat{\boldsymbol{r}} = \hat{\boldsymbol{u}}_{\mathrm{bk}}$, and the OCBS (Opposite-Circular-Basis Scattering) observable [9] is

$$S_{\mathrm{bk}}^{\mathrm{OCBS}} = \frac{-S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} + S_{\mathrm{bk},\phi}^{\mathrm{OCBS}}}{\sqrt{2}}. \tag{38}$$

As shown in [9], combining the two channels with this sign pattern cancels the off-diagonal Mishchenko elements $S_{12}(\pi) + S_{21}(\pi) = 0$ and produces a single, orientation-stable scalar observable.

8 Validation

8.1 Mie sphere

For a homogeneous sphere the exact analytical solution is the Mie series [8]. The reference implementation in `test/mie_reference.jl` provides C_{ext}, C_{sca}, C_{abs}, and the full complex amplitude matrix elements S_{11}, S_{22} at $\theta = 0, \pi$. The Phase-5 validation tests compare the VIEM solution to this Mie reference on a volume-equivalent sphere with $\lambda_0 = 4\,\mu\mathrm{m}$, $m_m = 1$, $m_p = 1.5 + 0.01i$, $r = 0.5\,\mu\mathrm{m}$ and mesh sizes $\ell_c \in \{0.30, 0.18, 0.12\}$. After the 2026-04-13 physics-convention switch (Sec. 7.1), both real and imaginary parts of the PCAS forward amplitude (37) agree with the Mie reference to $\sim 0.4\%$ at the coarsest mesh and $\sim 0.1\%$ at $\ell_c = 0.12$. All 1704 tests in the unit-test suite pass.

8.2 Oblate spheroid α-symmetry

A prolate or oblate spheroid has a continuous C_∞ rotation symmetry about its figure axis. When the orientation is parameterised by intrinsic ZYZ Euler angles with $\gamma = 0$, rotating the first angle α does not change the physical orientation of an axisymmetric particle in the particle frame; it only rotates the incident polarisation basis by α about $\hat{\boldsymbol{u}}_{\mathrm{inc}}$. For the CAS-v2 observables this implies the analytical α-expansion [9]

$$S_\theta(\alpha, \beta, \gamma{=}0) = A(\beta) + B(\beta)\,\mathrm{e}^{+2i\alpha}, \tag{39}$$

$$S_\phi(\alpha, \beta, \gamma{=}0) = A(\beta) - B(\beta)\,\mathrm{e}^{+2i\alpha}, \tag{40}$$

with

$$A(\beta) = \tfrac{1}{2}\big[S_\theta(0,\beta,0) + S_\phi(0,\beta,0)\big], \quad B(\beta) = \tfrac{1}{2}\big[S_\theta(0,\beta,0) - S_\phi(0,\beta,0)\big]. \tag{41}$$

The identity (39)–(40) is the exact basis of the α-analytical expansion that `block-DDA_Py` uses to reduce the orientation sweep to solves at $\alpha = 0$ only. Because VIEM after the physics-convention switch lives in the same time-convention as block-DDA_Py, the sign of the exponent is $+2i\alpha$ — identical to block-DDA_Py.

The benchmark script `benchmarks/cas_v2/spheroid_ar3.jl` builds an oblate spheroid with bc_ratio $= b/c = 3$, solves at $\beta = \pi/4$ for $\alpha \in \{0, \pi/8, \pi/4, 3\pi/8, \pi/2\}$, and compares the independently solved amplitudes to the analytical prediction (39)–(40). The maximum relative deviation is $\sim 5 \times 10^{-15}$, that is, machine precision for double-precision arithmetic.

The α-symmetry check verifies the internal consistency of block-VIEM.jl but does not confirm that the computed amplitudes themselves are correct. A direct cross-validation against block-DDA_Py on the same oblate spheroid is therefore performed by the driver pair under `benchmarks/cas_v2/dda_comparison/` (Python for DDA, Julia for VIEM, JSON as the exchange format):

- particle: oblate spheroid, $D_{ve} = 0.40\,\mu\text{m}$, bc_ratio = 3, $m_p = 1.5$ (non-absorbing);
- background: $m_m = 1$, $\lambda_0 = 0.638\,\mu\text{m}$ ($k_0 r_{ve} \approx 1.97$);
- orientations: two tilt angles, $(\alpha, \beta, \gamma) = (0, \pi/4, 0)$ and $(0, \pi/2, 0)$;
- discretisations: DDA uses $N_{\text{dip}} = 2236$ dipoles with dpl = 17; VIEM uses a half-SWG mesh with $\ell_c = 0.035\,\mu\text{m}$ yielding $N_{\text{dof}} = 8746$.

The PCAS forward amplitudes agree to within $\sim 3\%$ at both tilt angles:

Table 1: Cross-validation of VIEM against block-DDA_Py on a single oblate AR = 3 spheroid at two tilt angles. The relative error is $|\Delta S|/|S_{\text{DDA}}|$.

Observable	block-DDA_Py	block-VIEM.jl	rel. err.
$\beta = \pi/4$			
$S_{\text{fw},\theta}$	$+0.23514 + 0.09368i$	$+0.23143 + 0.09164i$	1.67%
$S_{\text{fw},\phi}$	$+0.23884 + 0.18561i$	$+0.23480 + 0.17937i$	2.46%
C_{ext}	$0.178\,19\,\mu\text{m}^2$	$0.172\,91\,\mu\text{m}^2$	2.96%
$\beta = \pi/2$			
$S_{\text{fw},\theta}$	$+0.23703 + 0.08742i$	$+0.23221 + 0.08536i$	2.08%
$S_{\text{fw},\phi}$	$+0.29578 + 0.21695i$	$+0.28980 + 0.20883i$	2.75%
C_{ext}	$0.194\,19\,\mu\text{m}^2$	$0.187\,69\,\mu\text{m}^2$	3.34%

The two tilt angles cover qualitatively different illumination geometries: at $\beta = \pi/4$ the spheroid is intermediate between flat-on and edge-on; at $\beta = \pi/2$ the figure axis is exactly perpendicular to $\hat{\boldsymbol{u}}_{\text{inc}}$ (edge-on), so the incident wave sees the largest geometrical cross section and the C_{ext} value is roughly 9% larger than at $\pi/4$. The fact that VIEM and DDA agree to the same few-percent level at both tilts — not just on the magnitude but on both the real and imaginary parts of $S_{\text{fw},\theta}$ and $S_{\text{fw},\phi}$ independently — demonstrates that the agreement is not an accidental near-coincidence at one orientation but a genuine convergence to the same physical scattering amplitudes. Both codes are discretisation-limited at these resolutions — the VIEM volume-equivalent radius recovered from the mesh is $r_{ve}^{\text{mesh}} = 0.199\,18\,\mu\text{m}$, already a 0.41% volumetric error on top of the half-SWG error budget, and DDA has its own stair-casing error of comparable size — so a $\sim 2\%$ disagreement between the two independent codes is the level of agreement one can expect at these mesh densities. The result confirms that block-VIEM.jl and block-DDA_Py produce the *same complex polarimetric amplitudes* on an axisymmetric non-spherical particle up to discretisation error, in addition to reproducing the Mie sphere limit to $\sim 0.4\%$ (Sec. 8.1).

8.3 Accuracy and computational cost vs. block-DDA_Py

Table 2 compares the C_{abs} error produced by block-VIEM.jl (half-SWG on a tetrahedral mesh) and block-DDA_Py (cubic-lattice DDA with the Chaumet–Rahmani dynamic polarisability) on the same homogeneous Mie sphere, $m_p = 1.5 + 0.01i$, $r = 1\,\mu\text{m}$, $\lambda_0 = 10\,\mu\text{m}$ (size parameter $x \approx 0.63$).

Table 2: Per-DoF accuracy comparison on a Mie sphere ($m_p = 1.5 + 0.01i$, $x \approx 0.63$). The VIEM solves use the half-SWG basis and Duffy volumetric quadrature; the DDA solves use block-BiCGSTAB on the Goodman–Draine–Flatau FFT operator.

Method	N_{dof}	C_{abs} rel. error
DDA, spacing $d = 0.14\,\mu\mathrm{m}$	4488	1.25%
half-SWG VIEM, $\ell_c = 0.50\,\mu\mathrm{m}$	589	0.21%
half-SWG VIEM, $\ell_c = 0.18\,\mu\mathrm{m}$	7868	0.15%

For this test block-VIEM.jl is roughly **10–12× more accurate per DoF** than block-DDA_Py: matching the 1.25% DDA error requires only ~ 400 half-SWG DoFs (versus ~ 4500 DDA dipoles), and at the same DoF count (~ 8000) the VIEM error is an order of magnitude smaller.

There are four mutually reinforcing reasons.

1. *Conforming geometry.* A tetrahedral mesh represents curved boundaries to $O(\ell_c^2)$, whereas a cubic lattice stair-cases them to $O(d)$. For Mie spheres this alone removes a geometric contribution to C_{abs} and C_{sca} that scales as d rather than d^2.

2. *Conforming vector space.* The SWG/half-SWG basis is $H(\mathrm{div})$-conforming, so the normal component of $\boldsymbol{D}$ is automatically continuous across tetrahedron faces — the physically correct boundary condition at a dielectric interface. Expanding $\boldsymbol{D}$ instead of $\boldsymbol{E}$ [1, Sec. II.B] further ensures that the polarisation jump at $\partial\Omega$ is represented exactly by the half-SWG functions on the boundary faces, with the charge-neutrality identity of Sec. 3.3. In DDA the induced polarisation is piecewise constant at the dipole centres, so the interface condition is only recovered in the limit $d \to 0$.

3. *Linear vector basis vs point dipoles.* Within each tetrahedron $\boldsymbol{D}$ is represented by up to four SWG functions, each linear in position and with a piecewise-constant divergence. DDA represents the polarisation by a single point dipole per element, i.e. a delta distribution localised at the lattice node. The tetrahedral moment approximation is therefore intrinsically one order higher, and in the far field the error per element scales as $k_0 h$ rather than $(k_0 d)^0$.

4. *Physical polarisability.* DDA requires a dynamic polarisability prescription (Clausius–Mossotti plus the $(2/3)(ka)^3$ radiative correction [6]) to cancel leading-order interaction errors, and the choice of prescription feeds back into the error. VIEM solves the integral equation directly; $\boldsymbol{D}$ is the physical variable, and no such tuning is required.

Against these accuracy advantages, block-VIEM.jl pays a higher per-DoF setup cost. Table 3 summarises the asymptotic cost and memory for a matrix-free MVP:

The central difference is that DDA has a strictly translation-invariant interaction matrix, so a single FFT on the doubled lattice captures the entire MVP exactly. VIEM has no such property: each basis function sees a different discrete support of neighbouring tetrahedra, and the AIM moment matching is only accurate for pairs whose separation is large compared with the element size. Near pairs must therefore be computed directly via Duffy quadrature and stored as a sparse precorrection matrix (Sec. 5.4). For typical settings, this near-field work dominates both the matrix assembly time and the per-iteration MVP cost at small to moderate N; only for very large meshes does the $N_g \log N_g$ FFT cost become comparable.

The overall trade-off is therefore: **VIEM spends more work per unknown (near-field Duffy integrals and precorrection), but uses an order of magnitude fewer unknowns to reach a given accuracy for high-contrast particles with curved boundaries.**

Table 3: Asymptotic cost model. N is the number of unknowns, N_g the number of AIM or FFT grid points (typically $N_g \sim 2N$ for AIM), $n_{\rm st}$ the AIM moment matching order, $n_{\rm it}$ the number of Krylov iterations, and L the number of particle orientations in a batch solve.

Quantity	block-DDA_Py	block-VIEM.jl (half-SWG + AIM)
Element geometry	cubic lattice	unstructured tetrahedra
Green kernel on grid	analytic dyadic, $\mathcal{O}(N_g)$	scalar + moment-matched dyadic, $\mathcal{O}(N_g)$
Near-field "precorrection"	none (exact on the lattice)	Duffy volumetric, $\mathcal{O}(N\, n_{\rm near})$
MVP cost	$\mathcal{O}(N_g \log N_g)$	$\mathcal{O}(N_g \log N_g + N\, n_{\rm near})$
Multi-orientation batch	block-BiCGSTAB on L RHS	shared LU (small N) or L independent BiCGSTA
Dominant memory	$\mathcal{O}(N_g)$ FFT kernel	$\mathcal{O}(N_g)$ FFT kernel $+\mathcal{O}(N\, n_{\rm near})$ precorr.

For sphere-like, low-contrast problems DDA's simplicity wins on wall-clock time; for the high-contrast iron-oxide aggregates and faceted gold nanoparticles targeted by this project, VIEM is the clear winner and the block-VIEM.jl $\ell_c = 0.50$ line of Table 2 is representative of production use.

9 Spheroid Sweep HDF5 Output

For parameter-sweep applications the forward scattering amplitudes on a five-dimensional grid are written to an HDF5 file whose layout is deliberately identical to that produced by `block-DDA_Py` [9], so the downstream consumer `PCAS_Bayes_APM_Nonspherical/lut_generation/build_spheroid_lut.py` reads VIEM output without modification. The schema is documented in `block-DDA_Py/.claude/spheroid_h5_` and summarised here for completeness.

9.1 Parameter grid

The five grid axes are stored at the root level of the file and shared across wavelength groups:

- `D_ve_grid` — volume-equivalent diameter D_{ve} [µm];
- `RI_real_grid` — real part of the particle refractive index, $\mathrm{Re}(m_p)$;
- `log_AR_grid` — base-10 logarithm of bc_ratio $= b/c$, following the `block-DDA_Py` convention where bc_ratio > 1 is oblate, < 1 is prolate, and $= 1$ is a sphere;
- `cos_theta_o_half_grid` — $\cos\theta_o$ on $[0, 1]$ (half-domain, extended to $[-1, 1]$ by the consumer using the $\theta_o \to \pi - \theta_o$ mirror symmetry);
- `phi_o_grid` — ϕ_o on $[0, \pi]$ (half-domain, extended to $[0, 2\pi]$ by the consumer using the ϕ_o-periodicity $S(\theta_o, \phi_o + \pi) = S(\theta_o, \phi_o)$).

All axes must be equidistant; `write_spheroid_sweep_h5` enforces this constraint and raises an error otherwise, because the downstream `NdimSpline_JAX` consumer assumes uniform grids.

9.2 Per-wavelength datasets

Results for each wavelength λ_0 are stored under a group `wl_<value>` (e.g. `wl_0p638` for $\lambda_0 = 0.638\,\mu\mathrm{m}$), with the attributes `wl_0` and `m_m`. Each group contains

- `S_fw_theta_re`, `S_fw_theta_im`, `S_fw_phi_re`, `S_fw_phi_im`: the real and imaginary parts of $S_{\mathrm{fw},\theta}$ and $S_{\mathrm{fw},\phi}$ on the shape $(N_{D_{ve}}, N_{RI}, N_{AR}, N_{u_{1/2}}, N_{\phi_o})$;

- `converged`: a boolean array of shape $(N_{D_{ve}}, N_{RI}, N_{AR})$ indicating successful solves. Non-converged cells carry `NaN+NaN`i in the four amplitude arrays across all $(N_{u_{1/2}}, N_{\phi_o})$ entries.

The ϕ_o axis corresponds exactly to the Euler angle α, and the populating code applies the analytical α-expansion (39)–(40) to the $\alpha = 0$ solves, so only $N_{u_{1/2}}$ independent VIEM solves are required per $(D_{ve}, \mathrm{Re}(m_p), \mathrm{AR})$ triple.

10 AIM-accelerated block-Krylov multi-orientation solve

For large orientation sweeps — orientation LUTs typically require $L = 10^2$–10^3 incidence directions per $(D_{ve}, \mathrm{Re}(m_p), \mathrm{AR})$ cell — reusing the same LU factorization of the dense impedance matrix (Sec. **??** style) becomes infeasible as soon as $N \gg 10^3$. For those problems `block-VIEM.jl` v0.2.0 exposes an AIM-accelerated *block-Krylov* path that solves

$$\mathbf{Z}\,\mathbf{X} = \mathbf{B}, \qquad \mathbf{X}, \mathbf{B} \in \mathbb{C}^{N \times L}, \tag{42}$$

with a single pre-built AIM operator and a single block iteration that acts on all L right-hand sides simultaneously.

Block MVP. The AIM operator supports the multi-RHS form

$$\mathrm{aim\_mvp}(\mathbf{X}) \;=\; \tfrac{1}{\epsilon_p}\mathbf{M}\mathbf{X} \;-\; \tfrac{\kappa}{\epsilon_{\mathrm{bg}}}\big[\mathbf{K}_{\mathrm{AIM}}(\mathbf{X}) + \mathbf{K}_{\mathrm{near}}\mathbf{X}\big] \;+\; \mathbf{K}_{\text{half-SWG}}\mathbf{X},$$

where $\mathbf{M}, \mathbf{K}_{\mathrm{near}}, \mathbf{K}_{\text{half-SWG}}$ are sparse and their matrix–matrix products cost $\mathcal{O}(\mathrm{nnz} \cdot L)$ in a single BLAS call. The far-field convolution $\mathbf{K}_{\mathrm{AIM}}(\mathbf{X})$ is applied column-by-column via the same FFT plans, so the FFT setup cost is amortized across all L columns at no extra memory.

Block BiCGSTAB. The default solver follows Tadano, Sakurai and Kuramashi [11] and is a direct port of `block-DDA_Py`'s `bl_bicgstab_jacobi_mvp_fft`, sans the Jacobi preconditioner (which is cheap for DDA because the dipole self-term is diagonal, but would require $\mathcal{O}(N)$ FFT probes for VIEM; the SWG mass matrix already normalizes the system adequately). Given $\mathbf{B} \in \mathbb{C}^{N \times L}$, set $\mathbf{X}_0 = \mathbf{0}$, $\mathbf{R}_0 = \mathbf{B}$, $\mathbf{P} = \mathbf{R}_0$, and $\widetilde{\mathbf{R}}_0 = \mathbf{R}_0$, then iterate

$$\begin{aligned}
\mathbf{V}_k &= \mathbf{A}\mathbf{P}_k, && (43)\\
\boldsymbol{\alpha}_k &= \big(\widetilde{\mathbf{R}}_0^H \mathbf{V}_k\big)^{-1}\big(\widetilde{\mathbf{R}}_0^H \mathbf{R}_k\big), && (44)\\
\mathbf{T}_k &= \mathbf{R}_k - \mathbf{V}_k \boldsymbol{\alpha}_k, && (45)\\
\mathbf{Z}_k &= \mathbf{A}\mathbf{T}_k, && (46)\\
\xi_k &= \frac{\mathrm{tr}\big(\mathbf{Z}_k^H \mathbf{T}_k\big)}{\mathrm{tr}\big(\mathbf{Z}_k^H \mathbf{Z}_k\big)}, && (47)\\
\mathbf{X}_{k+1} &= \mathbf{X}_k + \mathbf{P}_k \boldsymbol{\alpha}_k + \xi_k \mathbf{T}_k, && (48)\\
\mathbf{R}_{k+1} &= \mathbf{T}_k - \xi_k \mathbf{Z}_k, && (49)\\
\boldsymbol{\beta}_k &= -\big(\widetilde{\mathbf{R}}_0^H \mathbf{V}_k\big)^{-1}\big(\widetilde{\mathbf{R}}_0^H \mathbf{Z}_k\big), && (50)\\
\mathbf{P}_{k+1} &= \mathbf{R}_{k+1} + \big(\mathbf{P}_k - \xi_k \mathbf{V}_k\big)\boldsymbol{\beta}_k, && (51)
\end{aligned}$$

stopping once $\|\mathbf{R}_{k+1}\|_F / \|\mathbf{B}\|_F < \mathrm{tol}$. Each outer iteration performs exactly two block MVPs ($\mathbf{AP}$ and $\mathbf{AT}$) and four $L \times L$ Gram matrices, so the per-iteration cost is $\mathcal{O}\big(T_{\mathrm{MVP}} \cdot L + NL^2\big)$. Block BiCGSTAB can break down when the columns of $\mathbf{R}_k$ lose numerical independence; in practice this is rare for the physically distinct incidence directions of a CAS-v2 LUT, but the fallback below exists.

Block GMRES. The fallback is unrestarted Block GMRES [12]: a block-Arnoldi recurrence with thin QR factorization of each new block Krylov vector builds an orthonormal basis $\mathbf{V}_1, \mathbf{V}_2, \ldots$ of the block Krylov space $\mathcal{K}_k(\mathbf{A};\mathbf{B})$, together with an upper block-Hessenberg matrix $\underline{\mathbf{H}}_k$. At iteration k the approximate solution is $\mathbf{X}_k = \mathbf{X}_0 + [\mathbf{V}_1|\cdots|\mathbf{V}_k]\mathbf{Y}_k$, where $\mathbf{Y}_k$ minimizes $\|\underline{\mathbf{H}}_k\mathbf{Y} - \mathbf{E}_1\mathbf{\Lambda}\|_F$ with $\mathbf{\Lambda}$ the thin-QR factor of the initial residual. Unlike BiCGSTAB the memory grows linearly in the iteration count (k blocks of shape $N \times L$ are retained), so in practice Block GMRES is used as a robust fallback rather than a default.

Numerical verification. On an oblate AR= 3 spheroid ($D_{ve} = 0.40\,\mu\text{m}$, $m_p = 1.5$, $\lambda_0 = 0.638\,\mu\text{m}$, $N = 2009$ half-SWG DoFs) the two block solvers reach tolerance 10^{-8} in roughly 50 block iterations on $L = 5$ orientations, and their CAS-v2 forward amplitudes agree to $\sim 10^{-10}$ relative. Compared to the dense LU reference, the AIM block-Krylov observables agree to $\sim 0.8\%$, which is the AIM discretization error at the chosen pitch and not a Krylov convergence issue. The analytical $+2j\alpha$ spheroid symmetry (Sec. **??**) on the AIM block-BiCGSTAB solution holds at $\sim 5 \times 10^{-10}$, confirming that the block iteration preserves orientation symmetry to the solver tolerance.

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